Utsav D

AI & Machine Learning Enthusiast | B.Tech AI & Data Science Student

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Professional Summary

AI & **Data Science** undergraduate with hands-on experience in **Machine Learning**, **Deep Learning**, **Robotics**, and **Computer Vision**. Developed end-to-end AI solutions including a 6-DOF robotic manipulator simulation, a reinforcement learning-based process scheduler, and an AI-powered food recognition system. Proficient in **Python**, **PyTorch**, and **Flask**, with strong skills in model development, agentic AI systems, and multi-modal applications. Seeking a **Machine Learning** or **AI** internship to contribute to real-world, scalable intelligent systems.

Education

Amrita Vishwa Vidyapeetham | Bengaluru — Aug 2024 – May 2028
B.Tech in Artificial Intelligence & Data Science — *SGPA: 6.8/10*

Christ Junior College | Bengaluru — Jun 2022 – May 2024
Pre-University (PCMB) — *84%*

Projects

6-DOF Robotic Manipulator Simulation System | *Robotics & Simulation Engineer* — 2025 – Present

- Designed and implemented a comprehensive **6-Degree-of-Freedom** robotic manipulator simulation in **MATLAB**, incorporating advanced kinematic modelling.
- Programmed **volumetric path planning** algorithms to calculate collision-free trajectories in simulated 3D environments.
- Built interactive **3D visualizations** to validate forward and inverse kinematics, ensuring high spatial accuracy.
- *Technologies / Tools Used: MATLAB, Robotics Toolbox, Simulink*

Predictive Process Scheduling System | *Machine Learning & Systems Engineer* — 2025 – Present

- Built an intelligent process scheduling backend using a **Reinforcement Learning** model to predict burst times and optimize CPU scheduling.
- Resolved complex environment architecture issues and Python dependency conflicts in a **Flask** deployment pipeline.
- *Technologies / Tools Used: Python, Flask, OpenAI Gym, Stable-Baselines3*

AI-Powered Calorie & Nutrition Estimator | *Backend Developer & ML Engineer* — 2024 – 2025

- Engineered an end-to-end food analysis web app using **Google Gemini** and **Gradio**, identifying food items from images and estimating calorie/protein content in real time.
- Implemented image preprocessing, **multimodal API** integration, and structured nutritional insights with an interactive HTML-based UI.
- *Technologies / Tools Used: Python, Google Gemini API, Gradio, HTML*

ROBOVOX: Multi-Microphone Speaker Verification | *ML Researcher* — 2024 – 2025

- Designed a research-oriented **speaker verification** system using **PyTorch** with agentic microphone channel selection via reinforcement learning.
- Implemented **log-Mel feature extraction**, speaker embeddings, and EER-based evaluation pipelines with YAML configuration management.
- *Technologies / Tools Used: PyTorch, torchaudio, Gymnasium, Stable-Baselines3*

Skills

Programming Languages : Python, C, JavaScript, Flutter, SQL, HTML

AI / ML Frameworks : PyTorch, TensorFlow, Keras, scikit-learn, Stable-Baselines3, OpenAI Gym

Tools & Platforms : Flask, Gradio, Git, GitHub, Linux, MATLAB, Arduino, Google Colab

Databases : MySQL, PostgreSQL

Soft Skills : Communication, Teamwork, Quick Learner, Adaptability, Problem Solving

Languages : English (Professional), Kannada (Native), Hindi (Limited)

Additional Information

- **Certifications:** Udemy Course Completion Certificate (Machine Learning & Deep Learning)
- **Interests:** Open-Source Software, Powerlifting, Teaching, Bug Bounty, Fitness